# Flaxseed fed pork: n-3 fatty acid enrichment and contribution to dietary recommendations.

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The potential to increase n-3 fatty acid (FA) intake via flaxseed fed pork is underestimated when restricted to pure longissimus muscle, whereas a combination of muscle and adipose tissue is typically consumed. Presently, the FA content of pigs fed 0%, 5% and 10% dietary flaxseed for 11 weeks was measured in loin, picnic and butt primals (lean muscle with epimysium (L), L plus seam fat (LS), and LS plus 5 mm backfat (LSS)). The n-3 FA content necessary for an enrichment claim in Canada (300 mg/100 g serving) was exceeded in L from all primals when feeding 5% flaxseed, being 4 fold that of controls ($P<0.001$), with further enrichment from inclusion of associated adipose tissues ($P<0.001$). Increasing flaxseed feeding levels in combination with adipose tissue inclusion amplified total long chain n-3 FA ($P<0.05$), particularly 20:5n-3 and 22:5n-3. Flaxseed-fed n-3 FA enriched pork can contribute substantially to daily long chain n-3 FA intakes, particularly for societies with typically low seafood consumption. 